Convex Hulls in 3-Space

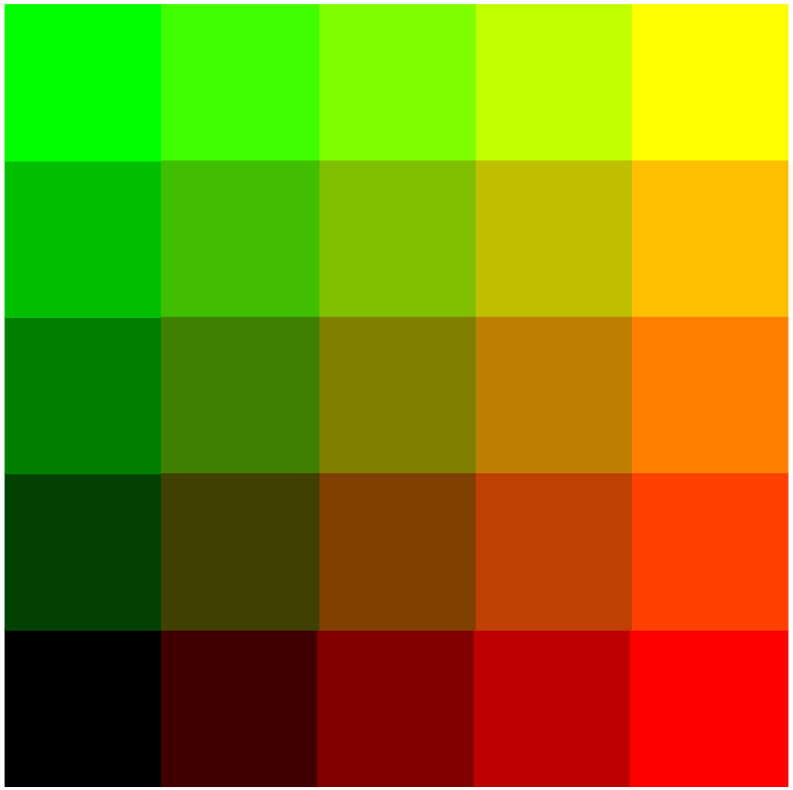
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   - definition of the convex hull
   - complexity of a convex polytope
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   - adding points in random order
   - visibility and horizon
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   - quickhull, incremental, and dynamic
   - convex hulls with scipy

MCS 481 Lecture 33
Computational Geometry
Jan Verschelde, 9 November 2026

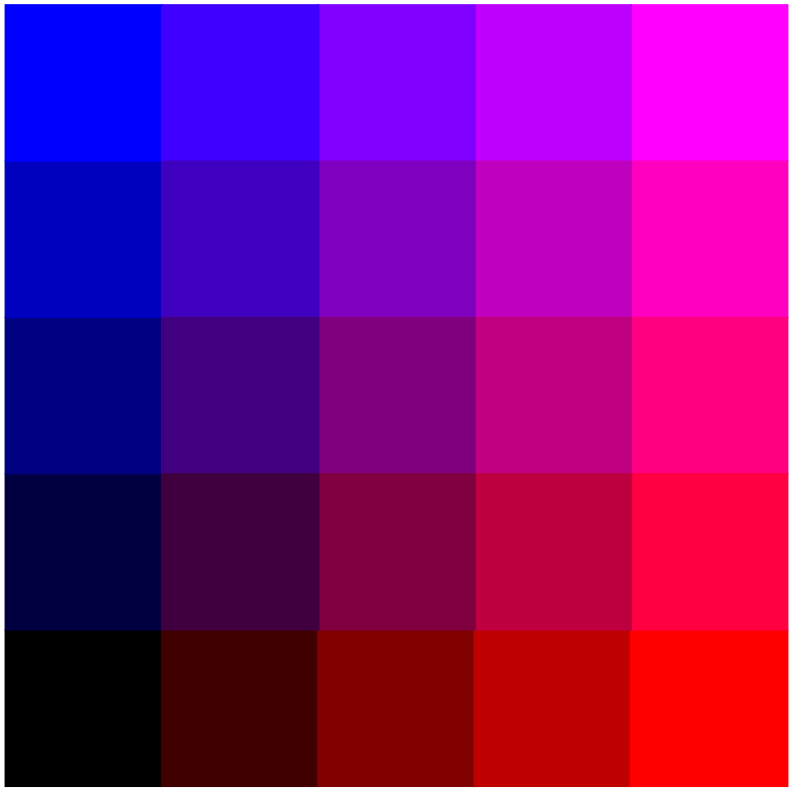
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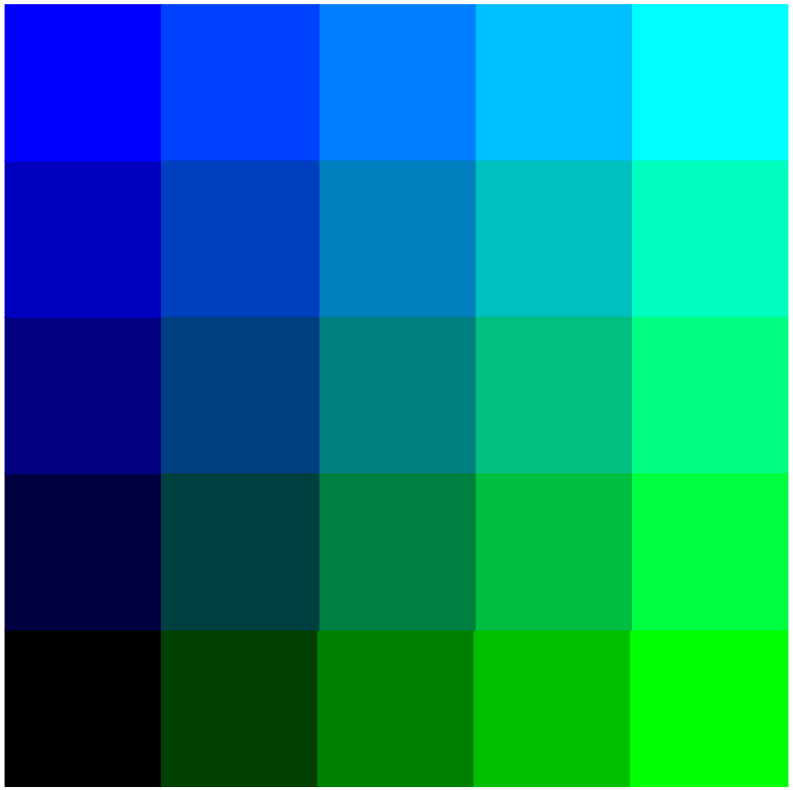
mixing red and green



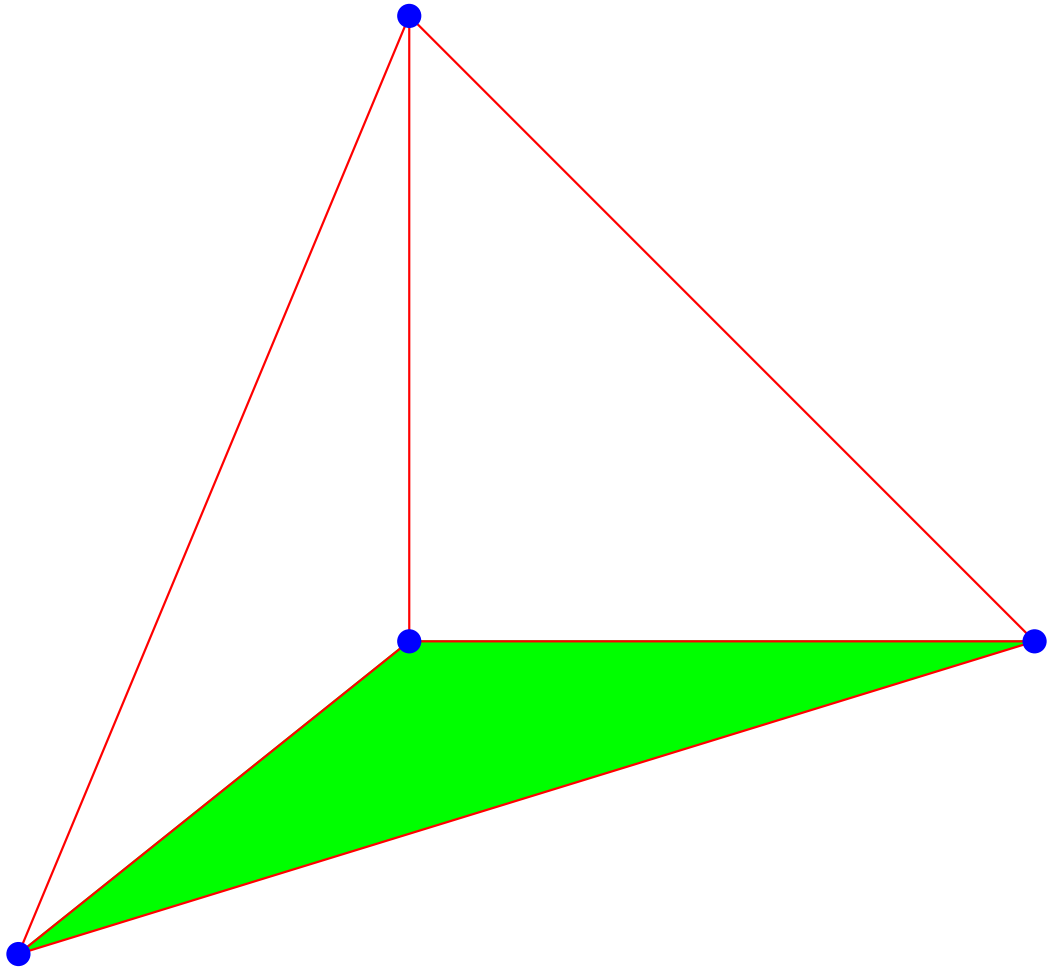
mixing red and blue



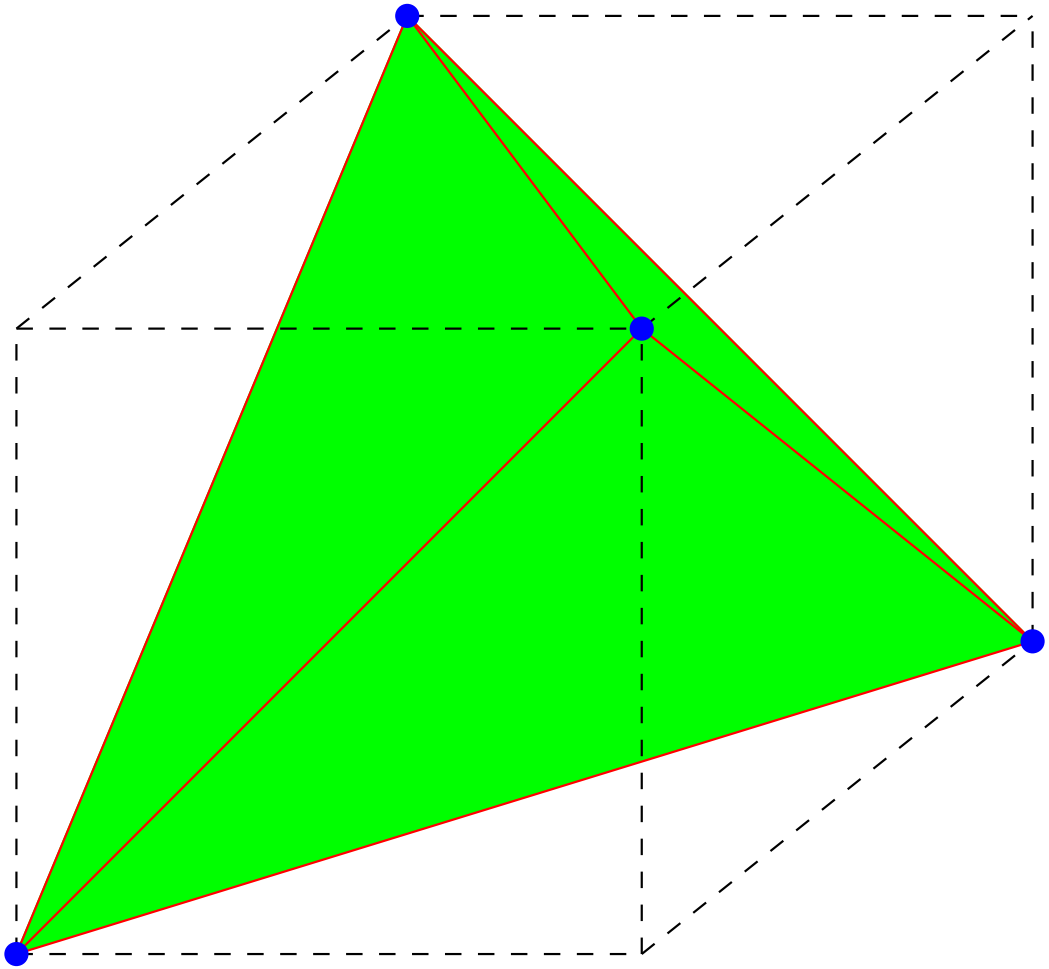
mixing green and blue



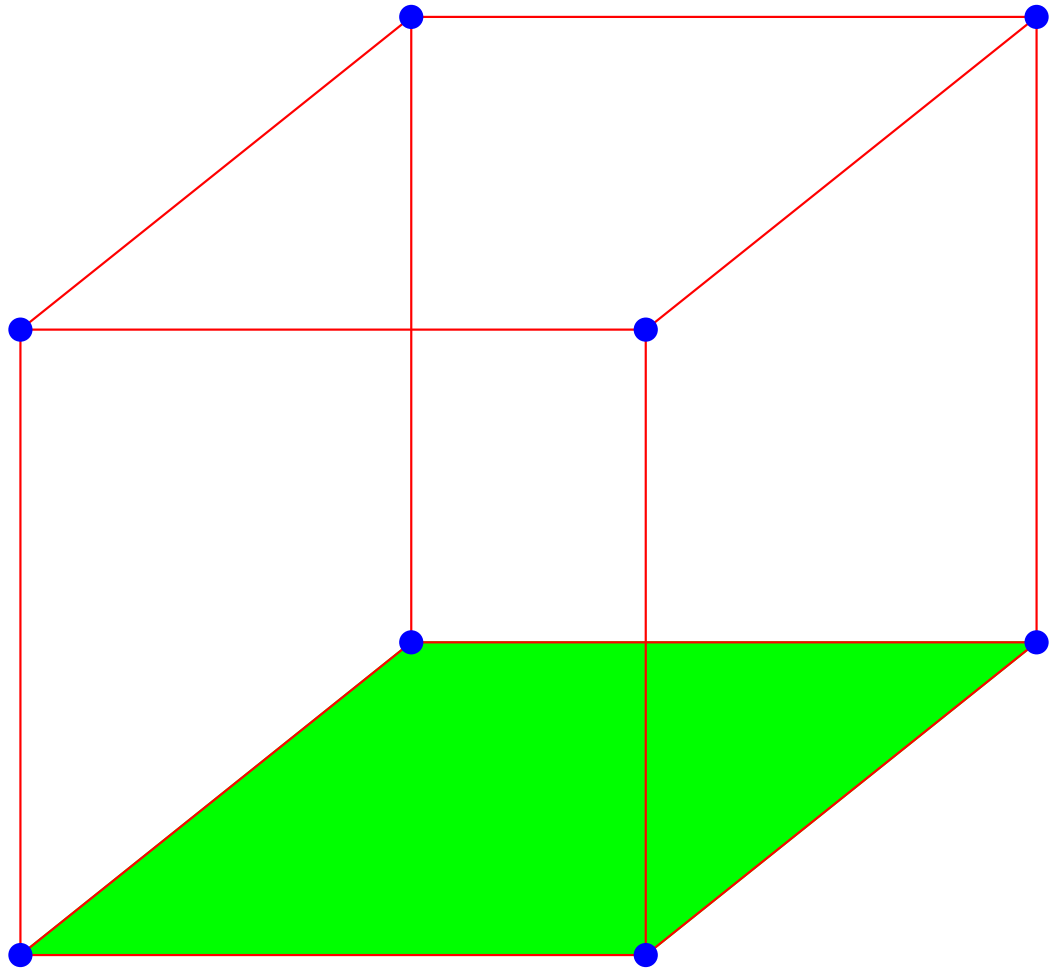
the simplex has 4 vertices, 6 edges, and 4 facets



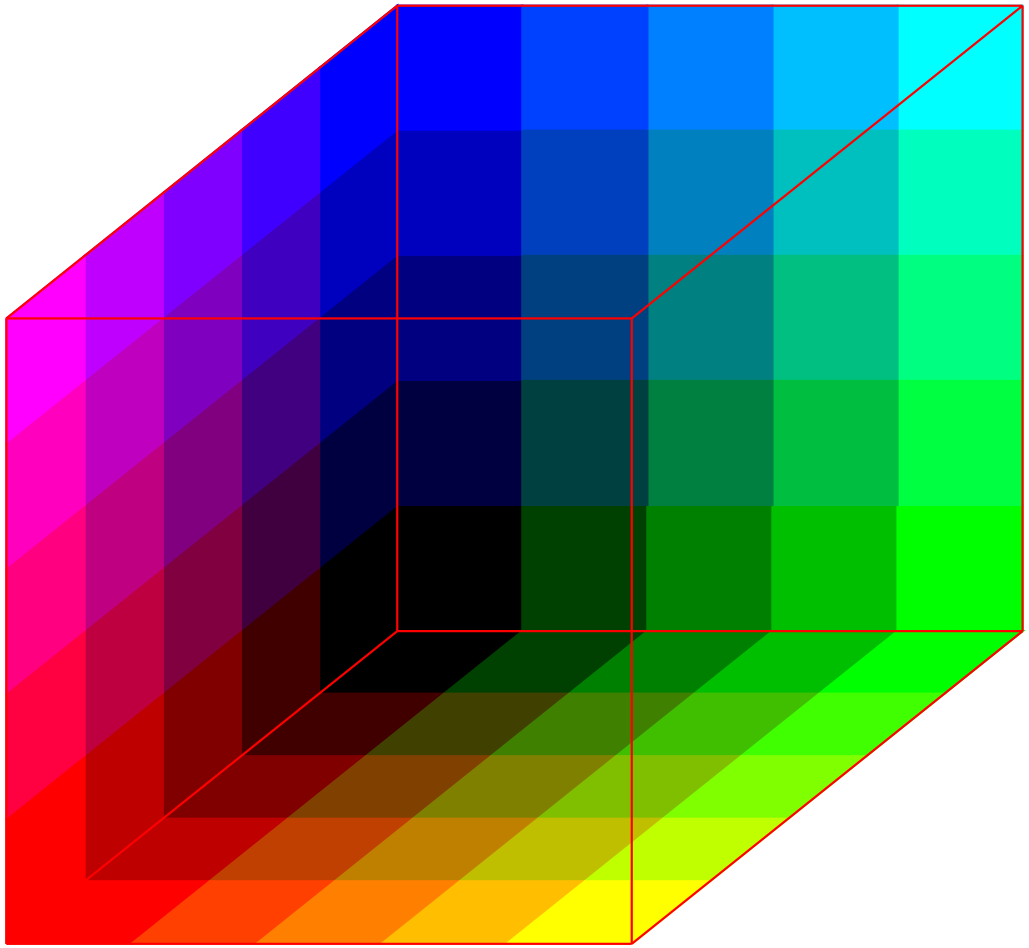
every facet is a triangle in a tetrahedron



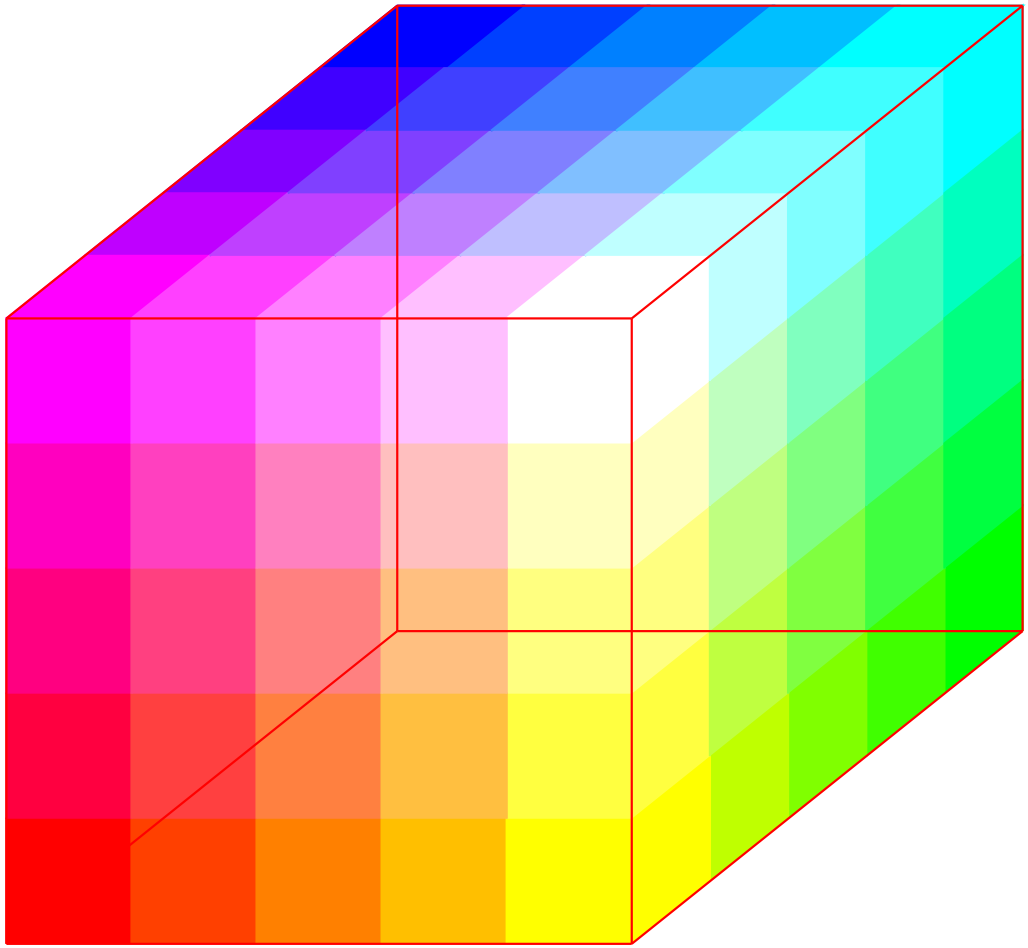
the cube has 8 vertices, 12 edges, and 6 facets



the three facets at the back of the color cube



the three facets at the front of the color cube



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convex combinations

Definition (convex combination)

Let $S = \{ p_1, p_2, \ldots, p_n \} \subset \mathbb{R}^3$, a set of n points in 3-space.
A *convex combination of the n points in S* is the point p, defined as

$$p = \sum_{i=1}^{n} \lambda_i p_i, \quad 1 = \sum_{i=1}^{n} \lambda_i, \quad \lambda_i \geq 0, i = 1, 2, \ldots, n.$$

The color cube consists of all convex combinations of the set

$$\{ \underbrace{(0,0,0)}_{\text{black}}, \underbrace{(1,0,0)}_{\text{red}}, \underbrace{(0,1,0)}_{\text{green}}, \underbrace{(0,0,1)}_{\text{blue}}, \underbrace{(1,1,0)}_{\text{yellow}}, \underbrace{(1,0,1)}_{\text{magenta}}, \underbrace{(0,1,1)}_{\text{cyan}}, \underbrace{(1,1,1)}_{\text{white}} \}.$$

the convex hull

Definition (convex hull)

Let $S = \{ p_1, p_2, \ldots, p_n \} \subset \mathbb{R}^3$, a set of n points in 3-space.
The *convex hull* $\mathrm{CH}(S)$ *of* S is the set of all convex combinations of the points in S.

Our problem is the following:

> Input: a set S of n points in 3-space.
> Output: $\mathrm{CH}(S)$, a doubly connected edge list,
> for all vertices, edges, and facets of the convex hull of S.

A convex polytope is the convex hull of its vertices.

How difficult is this problem?
$\rightarrow$ What is the size of the output to our problem?

For a set S of n points in the plane, $\mathrm{CH}(S)$ is $O(n)$.

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the complexity of a convex polytope

Theorem (number of vertices and edges of a convex polytope)

Let P be a convex polytope with n vertices.

1. *the number of edges of P is at most $3n - 6$, and*
2. *the number of facets of P is at most $2n - 4$.*

Are the bounds in the theorem sharp?

Let us verify. Consider our two examples.

1. The simplex has 4 vertices, 6 edges, and 4 facets.
2. The cube has 8 vertices, 12 edges, and 8 facets.

Corollary (complexity of a convex polytope)

The complexity of a convex polytope of n vertices is $O(n)$.

Euler's formula

Theorem (Euler's formula for a connected planar graph)

Let n_f, n_e, n_v respectively be the number of faces, the number of edges, and the number of vertices in a connected planar graph:

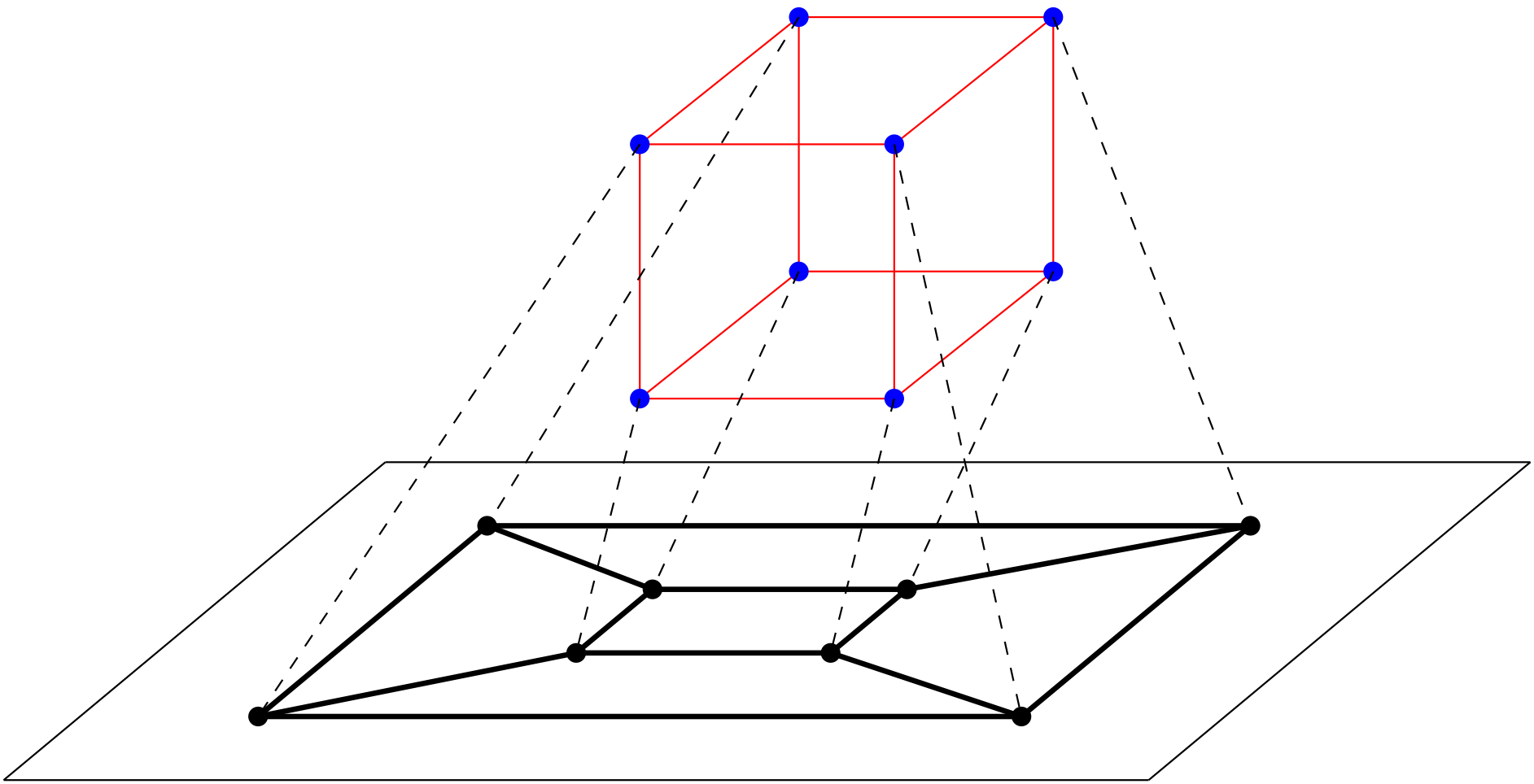
$$n_f - n_e + n_v = 2.$$

With every convex polytope P corresponds a planar graph $G = (V, E)$:

1. the vertices in V are the vertices of P,
2. $(v, w) \in E$ if there is an edge between v and w in P,
3. the facets of P are the faces in the graph.

One of the facets may corresponds to the unbounded face in G.

the planar graph of the cube



Exercise 1: Draw the planar graph of the simplex.

the planar graph of any convex polytope

Exercise 2: Why is the graph of a convex polytope planar?

application of Euler's formula

The number of vertices $n_v = n$, so: $n_f - n_e + n = 2$.

Every facet is incident to at least 3 edges: $2n_e \geq 3n_f$
and every edge is shared between exactly to facets.
If every facet is a triangle, then $2n_e = 3n_f$.

1. With $n_f \leq \frac{2}{3}n_e$, we eliminate n_f from Euler's formula:

$$2 = n_f - n_e + n \leq \frac{2}{3}n_e - n_e + n = -\frac{1}{3}n_e + n,$$

and we obtain

$$2 \leq -\frac{1}{3}n_e + n \quad \text{or equivalently} \quad 2 + \frac{1}{3}n_e \leq n.$$

Then $n_e \leq 3n - 6$ follows.

application of Euler's formula continued

The number of vertices $n_v = n$, so: $n_f - n_e + n = 2$.

Every facet is incident to at least 3 edges: $2n_e \geq 3n_f$
and every edge is shared between exactly to facets.
If every facet is a triangle, then $2n_e = 3n_f$.

2. With $n_e \geq \frac{3}{2}n_f$, we eliminate n_e from Euler's formula:

$$2 = n_f - n_e + n \geq n_f - \frac{3}{2}n_f + n = -\frac{1}{2}n_f + n,$$

and we obtain

$$2 \geq -\frac{1}{2}n_f + n \quad \text{or equivalently} \quad 2 + \frac{1}{2}n_f \leq n.$$

Then $n_f \leq 2n - 4$ follows.

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a randomized incremental algorithm

Algorithm ConvexHull(S)

Input: a set S of points in 3-space.
Output: $\mathrm{CH}(S)$, stored in a doubly connected edge list.

1. Δ = FindInitialTetrahedron(S)
2. if $\#\Delta < 4$ then all points in S are planar, return $\emptyset$
3. permute the remaining points $S_r = S \setminus \Delta = [p_1, p_2, \ldots, p_r]$
4. $\mathrm{CH}(S_0) = \{\Delta\}$
5. for k from 1 to r do
6. transform $\mathrm{CH}(S_{k-1})$ into $\mathrm{CH}(S_k)$, $S_k = [p_1, p_2, \ldots, p_k]$
7. return $\mathrm{CH}(S_r)$

finding the initial tetrahedron

1. Δ = FINDINITIALTETRAHEDRON(S)

Let q be an extremal point in S,
e.g.: q has the smallest x-coordinate, q is a vertex of $\mathrm{CH}(S)$.

For all points $p_i \in S \setminus \{q\}$, do row reduction on $p_i - q$.

After the row reduction, select the first three columns in the reduced matrix with nonzero pivot.

Then q and the points corresponding to the nonzero pivots are the points in the initial tetrahedron Δ.

Exercise 3: Run FINDINITIALTETRAHEDRON on the cube.

updating the convex hull with a point

Let p_k be the k-th point.
If $p_k \in \mathrm{CH}(S_{k-1})$, then do nothing,
otherwise update $\mathrm{CH}(S_{k-1})$, adding p_k.

How to decide whether $p_k \in \mathrm{CH}(S_{k-1})$?

We can maintain the half space representation of $\mathrm{CH}(S_{k-1})$:

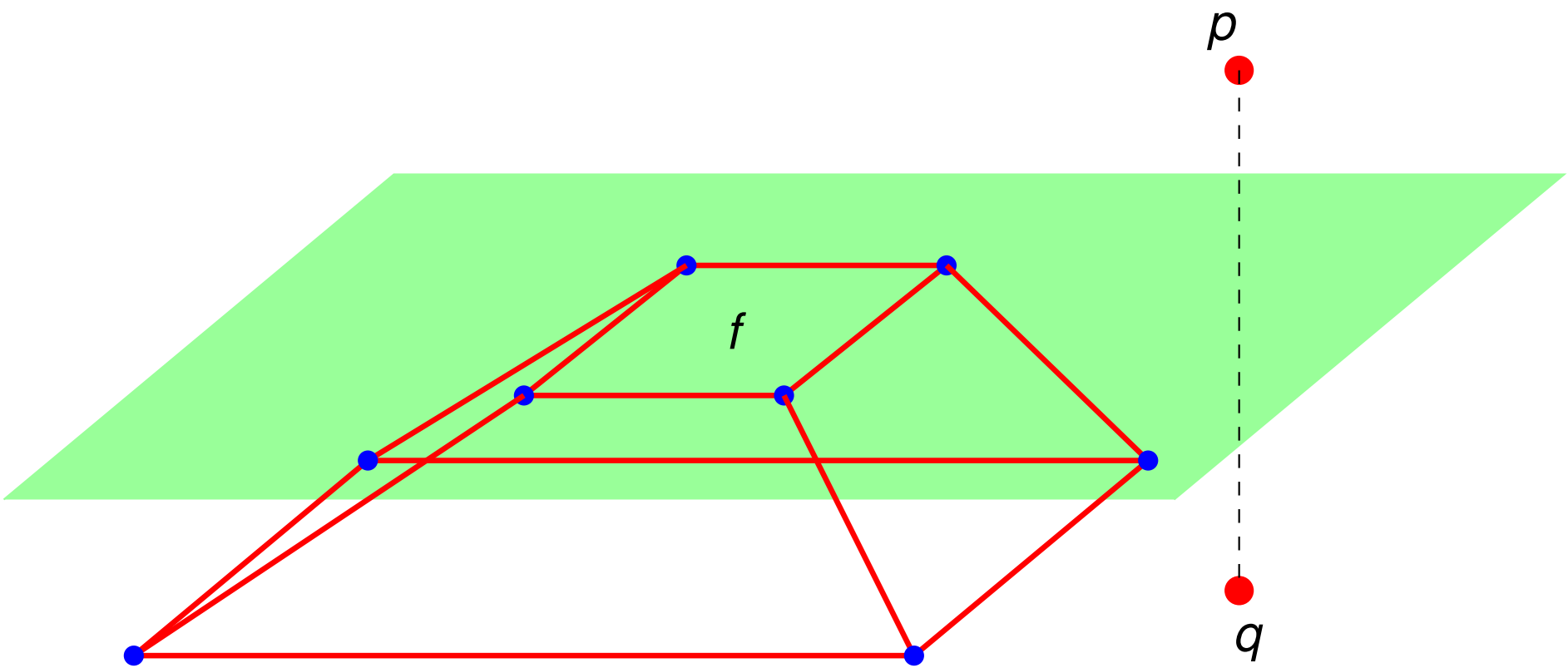
1. Through every facet passes a unique plane,
   defined by the vector perpendicular to the facet.
2. All points in $\mathrm{CH}(S_{k-1})$ lie in the same half space.
3. The half space is defined by one linear inequality.

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visibility of a facet from a point

Through every facet passes one plane and, moreover, all points of the polytope lie in the same half space defined by that plane.

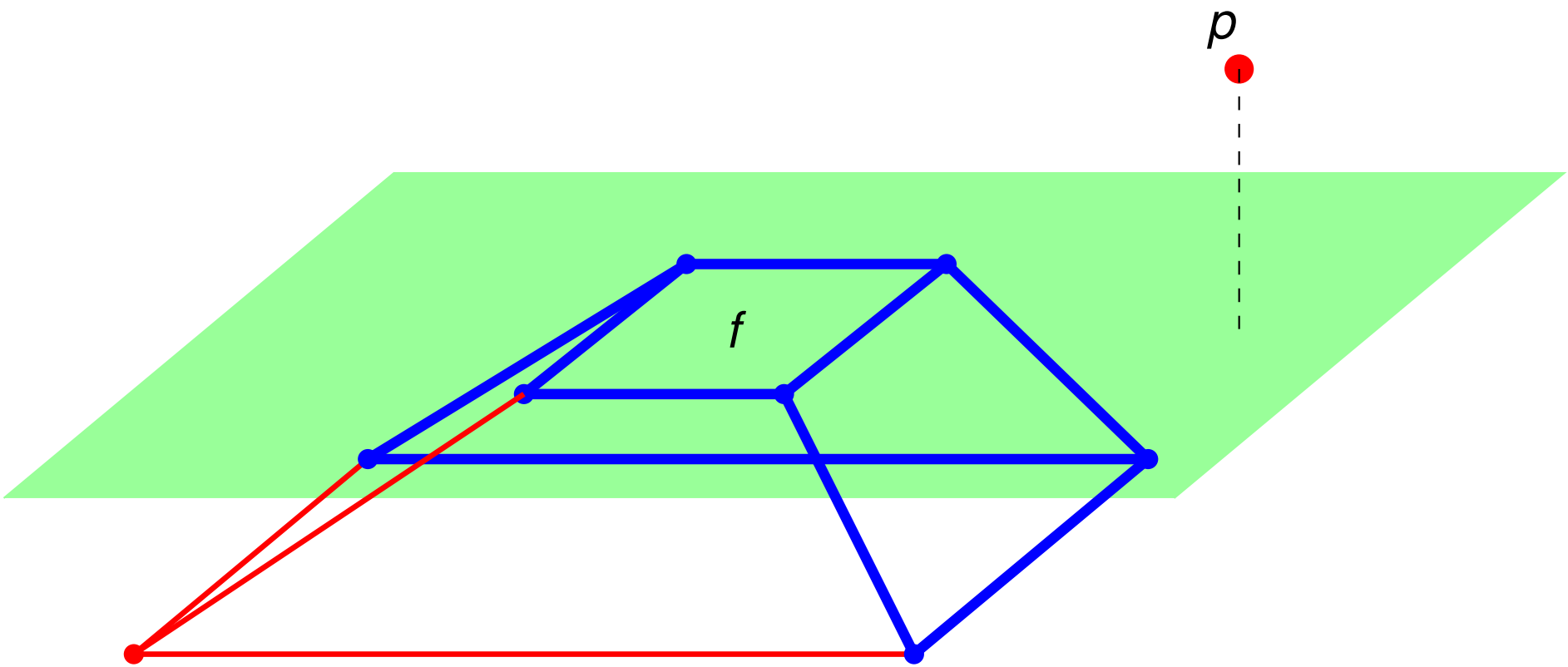


The facet f is visible from p, but not from q.

the horizon of a point

Definition (the horizon of a point)

Given a polytope P and a point p, *the horizon of p on P* is the border between the visible and the nonvisible facets of P.



cost of the algorithm

The complexity of $\mathrm{CH}(S_k)$ is $O(k)$.
$\Rightarrow$ finding the visible facets takes $O(k)$.

As we process n points, the cost of the algorithm is $O(n^2)$.

In the next lecture, we will see that we can do better than $O(n^2)$.

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convex hulls in CGAL and Qhull

Chapter 16 of the CGAL manual describes 3D convex hulls.

Three constructions of the convex hull are available:

1. Static with quickhull.
2. Incremental, for completeness and educational purposes.
3. Dynamic, uses a Delaunay triangulation.

The quickhull algorithm is implemented in the software Qhull.
The software Qhull is available as a standalone program.

C. Bradford Barber, David P. Dobkin, and Hannu Huhdanpaa.
The Quickhull algorithm for convex hulls.
ACM Trans. Math. Softw. 22(4):469-483, 1996.
Visit `http://www.qhull.org` for more information.

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convex hulls with scipy

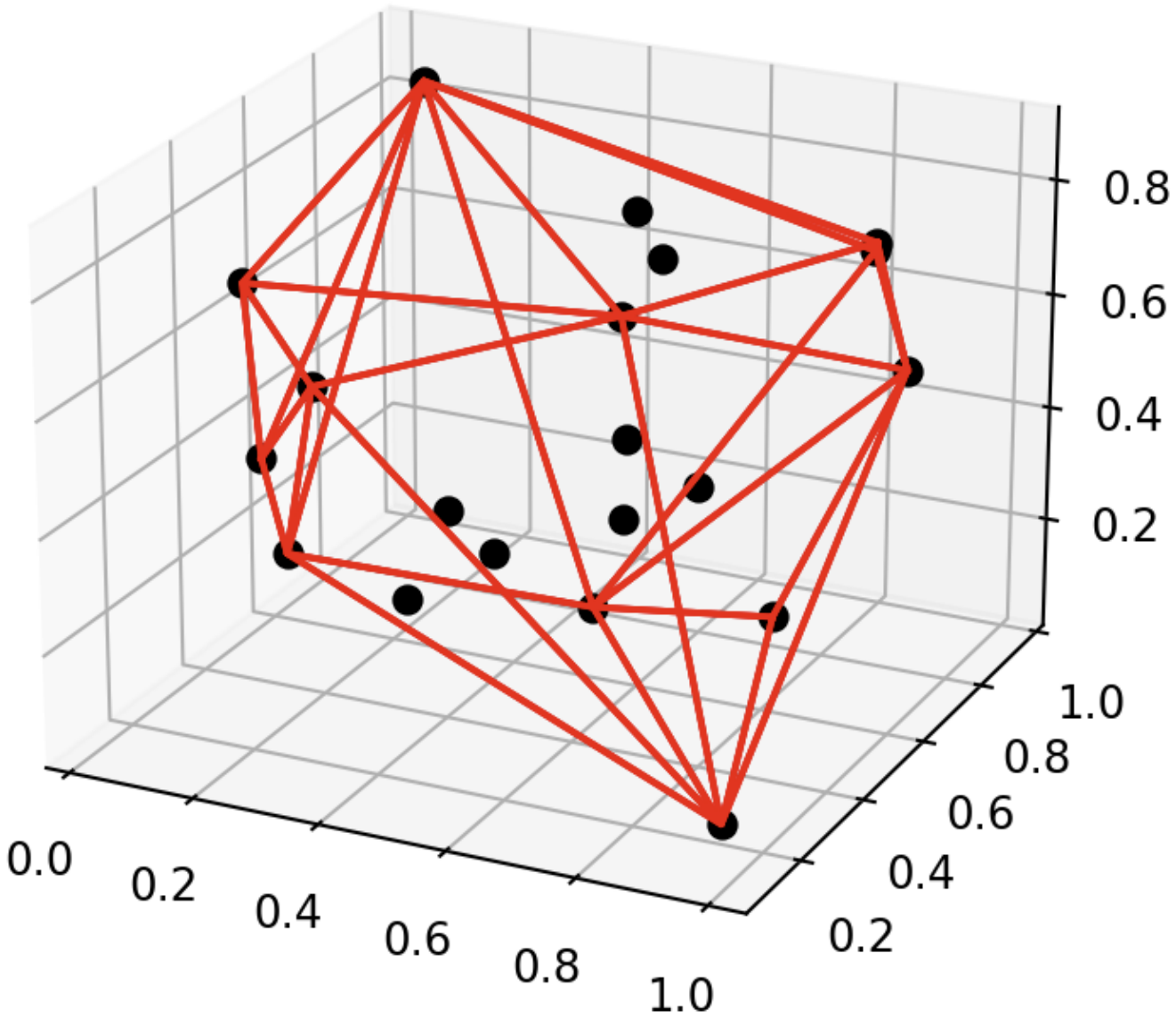
Let us compute the convex hull of 20 random points.

```
import numpy as np
from scipy.spatial import ConvexHull

points = np.random.rand(20, 3)
hull = ConvexHull(points)
print('the vertices :')
print(hull.vertices)
print('the facets :')
for simplex in hull.simplices:
    print(simplex)
```

For random points, we expect the convex polytope to be *simplicial*, that is: all facets are triangles.

plotting the facets with matplotlib



summary and exercises

The complexity of the convex hull is linear in the number n of points, allowing an $O(n \log(n))$ algorithm.

We started chapter 11 in the textbook.

Consider the following activities, listed below.

1. Write the solutions to exercises 1, 2, and 3.
2. Consult the CGAL documentation and example code on constructing convex hulls in 3-space.
3. Consider the exercises 1, 3, 4 in the textbook.